An overview of SJTUG mirror infrastructure

SJTUG

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Shanghai Jiao Tong University

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Current Status

Current Status

- Physical Resources
- Software Stack

Sync tools

- Daily Maintenance
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On-going Projects

- Renewed `lug` backend and top-level config schema
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Physical Resources

- Storage:
  - Local Storage @ Zhiyuan: 12.73 TiB, **93%** used
  - iSCSI @ Siyuan: 54.35 TiB, 56% used
  - jCloud S3: Bad performance to list-object 😾
    - Consider pre-caching S3 uploads in `mirror-intel` to mitigate this issue.

Software Stack

- Main repo
  - mirror-docker-unified: service orchestration via docker-compose; configuration management; templated code-gen for Caddyfile, S3 routing, etc.
- Sync tools
  - mirror-clone: general-purpose fetcher
  - mirror-intel: reverse proxy backend with smart cache
  - rsync-sjtug: rsync pulling to S3 bucket and HTTP serving from S3 bucket

  and helper scripts under `mirror-docker-unified:lug/worker-scripts`
  - lug: service backend, lightweight task scheduler for sync orchestration

Sync tools

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On-going Projects

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Daily Maintenance

- 检查服务状态
  - ~~清理又双叒叕满了的根分区~~
- 观测 mirror-requests issue 列表
  - 文档更新：在 mirror-docker-unified 或 portal 仓库下提交 PR 重定向至 MirrorZ Help

Daily Maintenance

- 观测 mirror-requests issue 列表
  - 新增镜像：评估可行后，根据对应同步方式提交 PR
    - 反代+缓存：在 mirror-intel 添加新的上游以及过滤规则，重新部署
    - Rsync：
      - worker-scripts/rsync-fetcher.sh
      - worker-scripts/rsync.sh (rsync-sjtug)
    - Git：
      - worker-scripts/git.sh
      - worker-scripts/github.sh
    - HTTP(S)：worker-scripts/mirror-clone-v2.sh

Daily Maintenance

- ~~以及清理同步工具的陈年 issue~~
  - mirror-intel#72: Return 404 for non-existent files instead of 403
  - mirror-docker-unified#467: Support secrets in lug config

On-going Projects

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On-going Projects

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Renewed `lug` backend and top-level config schema

Observability issues are the major bottleneck for the legacy `lug`.

- `lug` is designed to print

  (1) polling logs

  (2) sync script stdout/stderr

  directly to stdout. This results in no direct access to per-repo logs.

Renewed lug backend and top-level config schema

当前日志示例:

```
siyuan-lug  | time="2026-03-11T03:09:20Z" level=info msg="Start polling workers" event=poll_start manager=
siyuan-lug  | time="2026-03-11T03:09:20Z" level=info msg="Interval of w git/lean4-packages/batteries (3600 sec) elapsed, send it to pendingQueue" event=trigger_pending manager= target_worker_interval=3600 target_worker_name=git/lean4-packages/batteries
siyuan-lug  | time="2026-03-11T03:09:20Z" level=info msg="Interval of w ros (3600 sec) elapsed, send it to pendingQueue" event=trigger_pending manager= target_worker_interval=3600 target_worker_name=ros
siyuan-lug  | time="2026-03-11T03:09:20Z" level=info msg="Stop polling workers" event=poll_end manager=
siyuan-lug  | time="2026-03-11T03:09:30Z" level=info msg="Start polling workers" event=poll_start manager=
siyuan-lug  | time="2026-03-11T03:09:30Z" level=info msg="Interval of w git/homebrew-services.git (3601 sec) elapsed, send it to pendingQueue" event=trigger_pending manager= target_worker_interval=3601 target_worker_name=git/homebrew-services.git
siyuan-lug  | time="2026-03-11T03:09:30Z" level=info msg="Interval of w git/lean4-packages/lean4-cli (3600 sec) elapsed, send it to pendingQueue" event=trigger_pending manager= target_worker_interval=3600 target_worker_name=git/lean4-packages/lean4-cli
siyuan-lug  | time="2026-03-11T03:09:30Z" level=info msg="Stop polling workers" event=poll_end manager=
siyuan-lug  | time="2026-03-11T03:09:40Z" level=info msg="Start polling workers" event=poll_start manager=
siyuan-lug  | time="2026-03-11T03:09:40Z" level=info msg="Interval of w git/lean4-packages/ProofWidgets4 (3600 sec) elapsed, send it to pendingQueue" event=trigger_pending manager= target_worker_interval=3600 target_worker_name=git/lean4-packages/ProofWidgets4
```

Renewed `lug` backend and top-level config schema

What we want:

- Per-repo logging and metrics
- Better interaction and logging format with worker scripts
- Service integration, including Prometheus, Logstash, etc.

Monitoring

Currently `status.sjtug.org` provides only external status checks via HTTP requests.

It is beneficial to

- Per-repo access pattern & usage statistics
- Real-time traffic analysis

(`uptime-kuma` has a bad reputation on code quality thus we would like to consider alternatives.)

Infrastructure Cleanup

- `pytorch-wheels` still relies on the legacy version of `mirror-clone`.
- Migrate from the end-of-life Rust dependency `rusoto-s3` to `aws-sdk-s3`. (Already ported for `mirror-intel` but not deployed online, `mirror-clone` not touched yet.)
  - AWS does not provide any migration guide. 😾
- Deprecation policy for mirrors that are no longer maintained by upstream.

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Thanks for Listening!